

MATHEMATICS-III

Unit-4: Finite Fields

Comprehensive Study Notes

Unit-4: Finite Fields

1. Introduction to Finite Fields

1.1 What is a Field?

A **field** is a set F with two operations ($+$ and $\cdot$) that satisfies:

Axioms for Addition:

1. **Closure:** $a + b \in F$
2. **Associativity:** $(a+b)+c = a+(b+c)$
3. **Commutativity:** $a+b = b+a$
4. **Additive Identity:** $0 \in F$ such that $a+0 = a$
5. **Additive Inverse:** For each a , there exists $-a$ such that $a+(-a) = 0$

Axioms for Multiplication:

6. **Closure:** $a \cdot b \in F$
7. **Associativity:** $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
8. **Commutativity:** $a \cdot b = b \cdot a$
9. **Multiplicative Identity:** $1 \in F$ such that $a \cdot 1 = a$
10. **Multiplicative Inverse:** For each $a \neq 0$, there exists a^{-1} such that $a \cdot a^{-1} = 1$

Distributive Law:

11. **Distributivity:** $a \cdot (b+c) = a \cdot b + a \cdot c$

1.2 What is a Finite Field?

A **finite field** is a field with a **finite number** of elements. Also called a **Galois field** and denoted by $GF(q)$ or $\mathbb{F}_q$, where q is the number of elements.

Key Facts:

- The number of elements in a finite field is always of the form p^n , where p is prime and n is a positive integer.
- For every prime power p^n , there exists exactly one finite field $GF(p^n)$.

- $\text{GF}(p)$ is the field of integers modulo p .

1.3 Examples of Fields

Field	Elements	Operation	Is it a Field?
$\mathbb{Q}$	Rational numbers	$+, \times$	Yes
$\mathbb{R}$	Real numbers	$+, \times$	Yes
$\mathbb{C}$	Complex numbers	$+, \times$	Yes
$\mathbb{Z}$	Integers	$+, \times$	No (no multiplicative inverse for most)
$\mathbb{Z}_p$	Integers mod p (p prime)	$+, \times \bmod p$	Yes
$\mathbb{Z}_n$	Integers mod n (n composite)	$+, \times \bmod n$	No

2. GF(p) Fields

2.1 What is GF(p)?

GF(p) is the finite field with p elements, where p is a prime number. It consists of the integers $\{0, 1, 2, \dots, p-1\}$ with addition and multiplication modulo p .

$\text{GF}(5) = \{0, 1, 2, 3, 4\}$
 Addition modulo 5: $3 + 4 = 7 \equiv 2 \pmod{5}$
 Multiplication modulo 5: $3 \times 4 = 12 \equiv 2 \pmod{5}$
 $4 \times 4 = 16 \equiv 1 \pmod{5}$
 Inverse of 4 is 4 (since $4 \times 4 \equiv 1 \pmod{5}$)

2.2 Properties of GF(p)

Property	Description
Order	p (prime)
Characteristic	p (the smallest positive integer n such that $n \cdot 1 = 0$)
Non-zero Elements	$\{1, 2, \dots, p-1\}$ form a cyclic group under multiplication
Subfields	GF(p) has no proper subfields

2.3 Addition and Multiplication Tables

Addition Table for GF(5):

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2

4	4	0	1	2	3
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Multiplication Table for GF(5):

×	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

2.4 Finding Inverses in GF(p)

For any non-zero $a \in \text{GF}(p)$, find a^{-1} such that $a \times a^{-1} \equiv 1 \pmod{p}$.

Method: Extended Euclidean Algorithm or trial and error.

Example: Find inverse of 3 in GF(7).

Try: $3 \times 1 = 3$ (no)
 $3 \times 2 = 6 \equiv 6$ (no)
 $3 \times 3 = 9 \equiv 2$ (no)
 $3 \times 4 = 12 \equiv 5$ (no)
 $3 \times 5 = 15 \equiv 1$ (yes!)

Answer: $3^{-1} = 5$ in GF(7)

3. GF(p^n) Fields

3.1 What is GF(p^n)?

GF(p^n) is the finite field with p^n elements, where p is prime and $n \geq 1$. It is an extension field of GF(p).

Examples:

- $\text{GF}(4) = \text{GF}(2^2)$ — 4 elements
- $\text{GF}(8) = \text{GF}(2^3)$ — 8 elements
- $\text{GF}(9) = \text{GF}(3^2)$ — 9 elements
- $\text{GF}(16) = \text{GF}(2^4)$ — 16 elements

3.2 Construction of GF(p^n)

GF(p^n) is constructed as the set of all polynomials of degree less than n with coefficients in GF(p), modulo an irreducible polynomial of degree n .

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GF(4) = GF(2^2)
Elements: {0, 1, x, x+1}
Using irreducible polynomial: x^2 + x + 1 = 0

Multiplication:
x * x = x^2 = x + 1 (since x^2 + x + 1 = 0)
x * (x+1) = x^2 + x = (x+1) + x = 1
(x+1) * (x+1) = x^2 + 1 = x + 1 + 1 = x
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3.3 Elements of GF(p^n)

GF(p^n) has p^n elements, representable as:

- **Polynomial:** $a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1x + a_0$
- **Vector:** $(a_{n-1}, a_{n-2}, \dots, a_1, a_0)$
- **Exponential:** $0, \alpha^0, \alpha^1, \dots, \alpha^{p^n-2}$

3.4 GF(2^n) — Important for Cryptography

$GF(2^n)$ is particularly important in cryptography and error-correcting codes because operations are efficient in hardware.

Characteristics:

- Addition is XOR
- Multiplication is polynomial multiplication modulo irreducible polynomial
- $GF(2^8)$ is used in AES (Advanced Encryption Standard)

4. Polynomials and Their Operations Over GF(2)

4.1 Polynomials over GF(2)

A polynomial over GF(2) has coefficients that are either 0 or 1.

Examples:

$$\begin{aligned}f(x) &= x^3 + x^2 + 1 \\g(x) &= x^2 + x \\h(x) &= x^4 + x + 1\end{aligned}$$

4.2 Addition over GF(2)

Addition of polynomials over GF(2) is done by **XORing** coefficients.

Example:

$$\begin{aligned}f(x) &= x^3 + x^2 + 1 \\g(x) &= x^2 + x \\f(x) + g(x) &= (x^3 + x^2 + 1) + (x^2 + x) \\&= x^3 + (1+1)x^2 + x + 1 \\&= x^3 + 0x^2 + x + 1 \\&= x^3 + x + 1\end{aligned}$$

4.3 Multiplication over GF(2)

Multiplication is polynomial multiplication followed by reducing modulo an irreducible polynomial.

Example: Multiply $(x+1)$ and (x^2+1) over GF(2)

$$(x+1)(x^2+1) = x^3 + x^2 + x + 1$$

4.4 Irreducible Polynomials

An irreducible polynomial cannot be factored into polynomials of lower degree.

Irreducible Polynomials over GF(2):

Degree	Irreducible Polynomials
1	$x, x+1$
2	$x^2 + x + 1$
3	$x^3 + x + 1, x^3 + x^2 + 1$
4	$x^4 + x + 1, x^4 + x^3 + 1, x^4 + x^3 + x^2 + x + 1$
8	$x^8 + x^4 + x^3 + x + 1$ (Used in AES)

5. Operations in $GF(2^n)$

5.1 Polynomials as Field Elements

In $GF(2^n)$, elements are polynomials of degree less than n over $GF(2)$.

Example: $GF(8) = GF(2^3)$

Elements: $\{0, 1, x, x+1, x^2, x^2+1, x^2+x, x^2+x+1\}$

5.2 Addition in $GF(2^n)$

Addition is performed by XORing coefficients.

Example: In $GF(8)$ with irreducible polynomial $x^3 + x + 1$:

$$(x^2 + x) + (x^2 + 1) = (x^2 + x^2) + x + 1 = x + 1$$

5.3 Multiplication in $GF(2^n)$

Steps:

1. Multiply polynomials
2. Reduce modulo irreducible polynomial

Example: In $GF(8)$ with irreducible polynomial $x^3 + x + 1$, multiply $(x+1)$ and (x^2+1) :

$$\begin{aligned} \text{Step 1: } (x+1)(x^2+1) &= x^3 + x^2 + x + 1 \\ \text{Step 2: Reduce modulo } x^3 + x + 1 & \\ x^3 &\equiv x + 1 \\ x^3 + x^2 + x + 1 &\equiv (x+1) + x^2 + x + 1 \\ &= x^2 + (x+x) + (1+1) \\ &= x^2 + 0 + 0 \\ &= x^2 \end{aligned}$$

Answer: $(x+1)(x^2+1) = x^2$

5.4 Multiplicative Inverse in $\text{GF}(2^n)$

The inverse of a non-zero element can be found using the Extended Euclidean Algorithm.

Example: Find inverse of x in $\text{GF}(8)$ with irreducible polynomial $x^3 + x + 1$

$$x^3 + x + 1 = 0$$

$$x^3 = x + 1$$

We need y such that $x * y \equiv 1$

$$\text{Try: } x * (x^2 + 1) = x^3 + x = (x+1) + x = 1$$

Answer: $x^{-1} = x^2 + 1$

6. HPU Important Questions

Short Answer Questions (2–5 Marks)

1. What is a finite field?
2. Define $GF(p)$ field.
3. Define $GF(p^n)$ field.
4. What is an irreducible polynomial?
5. What is the characteristic of a field?
6. What is the order of a finite field?
7. How many elements does $GF(2^3)$ have?
8. What is the difference between $GF(p)$ and $GF(p^n)$?
9. Write the irreducible polynomial for $GF(8)$.
10. What is polynomial addition in $GF(2)$?

Long Answer Questions (10 Marks)

1. Define a finite field. Explain $GF(p)$ with examples.
2. Explain $GF(p^n)$ fields with construction methods.
3. Explain polynomial operations over $GF(2)$ and $GF(2^n)$.
4. What is an irreducible polynomial? List irreducible polynomials over $GF(2)$.
5. Multiply (x^2+1) and $(x+1)$ in $GF(8)$ with irreducible polynomial x^3+x+1 .
6. Find the inverse of $x+1$ in $GF(8)$ with irreducible polynomial x^3+x+1 .

Objective Questions (1 Mark)

1. The number of elements in a finite field is:
(a) p (b) p^2 (c) p^n (d) 2^n
2. $GF(5)$ has:
(a) 3 elements (b) 4 elements (c) 5 elements (d) 6 elements

Answer: (c)

3. $GF(2^3)$ has:
(a) 2 elements (b) 4 elements (c) 6 elements (d) 8 elements

Answer: (d)

4. $x^2 + x + 1$ is irreducible over:

- (a) GF(2) (b) GF(3) (c) Both (d) None

Answer: (a)

5. In GF(2), addition is:

- (a) XOR (b) AND (c) OR (d) NOT

Answer: (a)

6. The characteristic of GF(2^n) is:

- (a) 2 (b) 2^n (c) n (d) 1

Answer: (a)

Quick Revision

Finite Field Summary

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Finite Fields (GF)
|
+-- GF(p) - Prime Fields
|   - p elements
|   - p prime
|   - Integers mod p
|
+-- GF(p^n) - Extension Fields
    - p^n elements
    - Polynomials mod irreducible polynomial
    - GF(2^n) used in cryptography
  
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Key Facts

Field	Elements	Characteristic
GF(2)	{0, 1}	2
GF(3)	{0, 1, 2}	3
GF(5)	{0, 1, 2, 3, 4}	5
GF(4)	{0, 1, α , α^2 }	2
GF(8)	0, 1, x , $x+1$, x^2 , x^2+1 , x^2+x , x^2+x+1	2

Irreducible Polynomials over GF(2)

Degree	Polynomials
1	x , $x+1$
2	x^2+x+1

3	x^3+x+1, x^3+x^2+1
4	$x^4+x+1, x^4+x^3+1, x^4+x^3+x^2+x+1$

End of Unit-IV Notes — MATHEMATICS-III (HPU BCA)